

Sajeevan Veeriah

Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer

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Professional profile

Mechatronics engineer working across robotics, industrial controls, embedded electronics and engineering software. Combines automotive validation and manufacturing experience with independent client delivery, from requirements and design through integration, testing and handover. Recent work spans weighbridge controllers, vehicle-position sensing, business automation and community platforms.

Technical capabilities

Automation and controls: PLC, Siemens TIA Portal, HMI/SCADA, iFIX, PVI+, drives, MES, Modbus, PROFINET, FAT/SAT and GMP documentation.

Robotics and embedded: ROS 2, Nav2, SLAM, EKF, Gazebo, ESP32, STM32, FreeRTOS, C/C++, CAN, UART, I2C, SPI and BLE.

Software and AI: Python, MATLAB/Simulink, TypeScript, React, SQL, Git, CI/CD, OpenCV, scikit-learn and local AI retrieval workflows.

Design and test: SolidWorks, Fusion 360, KiCad, Altium, PCB/enclosure design, CANoe, CANalyzer, Linux, LoRaWAN, MQTT, InfluxDB and Grafana.

Professional experience

Independent Consultant Engineer

SV Tech Solutions | Jun 2026 - present

- Develop client systems for Stan Wootton Locksmiths and Gendio, covering requirements, electronics, embedded firmware, application software and release documentation.
- Deliver SWL pricing and inventory software with reviewed ServiceM8 imports, deterministic pricing, exception handling and rollback reports.
- Design Gendio controller and serial-adaptor PCBs; develop display firmware, browser configuration and vehicle-detection/distance-measurement integration. Coordinate fabrication reviews and maintain software verification records.

Automation and Controls Engineer

JAG Process Solutions | Jan 2026 - Jun 2026

- Integrated PLC, HMI/SCADA, field devices, drives and production data for pharmaceutical, biotechnology and food manufacturing.
- Migrated validated iFIX content to PVI+; supported FAT/SAT, commissioning, GMP handover and MiR mobile-robot fault tracing.

Product Development Test Engineer - Contract

Ford Motor Company via Invenio | Oct 2025 - Jan 2026

- Tested vehicle-software integration and ADAS through feature-vehicle, breadboard and regression testing. Instrumented vehicles and analysed CAN/CAN FD traces in CANoe and CANalyzer to isolate faults and document readiness.

Technical Officer and Quality Assurance

ABMARC | Jul 2024 - Aug 2025

- Conducted emissions, EV/PHEV range and vehicle-system testing using calibrated instrumentation; maintained traceable test, safety and certification records.

Consultant Engineer, IoT and Projects Administrator

DuxTel | Feb 2024 - Aug 2024

- Built ESP32 and LoRaWAN AU915 telemetry through ChirpStack, MQTT, InfluxDB and Grafana; integrated CAN/GPS, custom PCB work, MikroTik connectivity and Linux services for field trials and handover.

Selected engineering projects

Gendio controller and vehicle positioning Designed a four-layer ESP32-S3 controller, serial adapter, display firmware and local configuration interface. Extended the system with radar-based vehicle-position sensing, gateway integration and stale-data handling. Prepared CAD/manufacturing files and software checks; assembled-system and field validation remain pending.

SWL pricing and inventory control Sole engineer across React/TypeScript, Tauri/Rust, SQLite, pricing rules, automated tests and release packaging. Built operator-reviewed ServiceM8 imports with change, exception, rollback and audit reports.

NDCC platform and Dino Coach Develop and maintain the cricket club website, committee CMS and fantasy-cricket pilot using Next.js, Supabase and Stripe. Work includes registration, team management, budget tracking, payments, meal ordering and operational support.

SPC-001 waterless solar-panel cleaner Developed a modular tracked cleaning system covering mechanical CAD, electronics, supervisory control and coverage planning. Produced an independent research paper examining contact mechanics, traction and energy, with analytical results distinguished from physical performance.

Deadline-aware robot runtime assurance Developed a simulation study combining sensor uncertainty, measurement age and command delay in a braking supervisor. Built fault-injection benchmarks and compared recovery strategies under bounded sensing and timing assumptions.

Potato-processing automation simulation Built an independent browser-based process simulation with industrial control and operator-interface concepts, informed by Farm Frites discussions. Demonstration uses simulated data; no factory implementation is claimed.

Autonomous navigation rover Built a differential-drive platform with LiDAR and IMU sensing; integrated SLAM, EKF localisation and Nav2 in C++/Python. Checked transforms, paths and recovery in Gazebo and RViz; autonomy validated in simulation.

Ataxia assessment device Designed an ESP32 prototype, Altium PCB and enclosure with four Hall-effect channels and 100 Hz acquisition. Added Bluetooth display and reporting; used MATLAB to assess accuracy, reversals, drift and temperature behaviour.

VeerAI and engineering learning tools Built a local AI system with document ingestion, retrieval, memory, tools and evaluation. Developed Engineering Mastery Lab with technical learning workflows, engineering calculations and software checks.

Earlier manufacturing and quality experience

IDL Australia | Aug 2022 - Feb 2024 Production worker, supervisor and commissioning support to Sep 2023; cellar hand thereafter. Coordinated output, changeovers and handovers; supported canning-line commissioning, fault recovery, sampling and traceability.

Carbon Revolution | Sep 2021 - Dec 2021 Production line operator and commissioning support. Operated carbon-fibre wheel equipment and supported quality checks, equipment trials and first-level fault recovery.

Thornton Engineering Australia | Mar 2018 - Mar 2020 Undergraduate Quality Engineer. Supported drawing review, ITPs, MDRs, material traceability and AS/NZS compliance in structural-steel fabrication.

Education and professional membership

Bachelor of Mechatronics Engineering Honours Deakin University | Distinction | 2025

Higher National Diploma in Mechatronics, Robotics and Automation Engineering Cardiff Metropolitan University | Distinction | 2016

Member, Engineers Australia. Languages: English, Tamil and Sinhala. Community leadership, engineering mentoring and peer support.